

Received July 11, 2021, accepted July 18, 2021, date of publication July 26, 2021, date of current version August 20, 2021.

Digital Object Identifier 10.1109/ACCESS.2021.3099795

A Stacking Ensemble Prediction Model for the Occurrences of Major Adverse Cardiovascular Events in Patients With Acute Coronary Syndrome on Imbalanced Data

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This work was supported in part by the Korea Institute for Advancement of Technology (KIAT) Grant through the Korean Government by the Ministry of Trade, Industry and Energy (MOTIE) under Grant N0002429, in part by the Basic Science Research Program through the National Research Foundation of Korea (NRF) by the Ministry of Education under Grant 2017R1D1A1A02018718, and in part by the Ministry of Science and ICT (MSIT), South Korea, through the Grand Information Technology Research Center Support Program supervised by the Institute for Information and Communications Technology Planning and Evaluation (IITP) under Grant IITP-2021-2020-0-01462.

ABSTRACT The major adverse cardiovascular events (MACE) often occur with high morbidity and mortality globally. It is very important to predict the MACE occurrences accurately in patients with acute coronary syndrome (ACS). Therefore, this paper proposes a stacking ensemble model for the prediction of MACE occurrences in patients with ACS at early stage. Our research contents are summarized as follows. First, we use the Korea Acute Myocardial Infarction Registry National Institutes of Health (KAMIR-NIH) dataset, and our experimental data are extracted from the raw data and preprocessed. Second, we apply three data sampling approaches, such as borderline synthetic minority oversampling technique (Borderline-SMOTE1), cluster centroids undersampling, and synthetic minority oversampling techniques (SMOTE) plus Tomek Links (SMOTETomek) hybrid technique, to solve the class imbalance problem. Third, to develop a stacking ensemble prediction model for the occurrences of MACE, we apply seven widely used machine learning algorithms, such as logistic regression (LR), support vector machine (SVM), K-Nearest Neighbors (KNN), decision tree (DT), random forest (RF), extreme gradient boosting (XGBoost), and adaptive boosting (AdaBoost), as base learners. Fourth, the performance of the proposed stacking ensemble model is compared with the seven base learners using the three data sampling techniques. In the result, the proposed stacking ensemble model with the SMOTETomek shows the best performance with accuracy of 0.9862, precision 0.9976, recall 0.975, F1-score 0.9862, g-mean 0.9863, and AUC 0.9863 and provided a better solution for imbalanced dataset. Consequently, our finding is that the proposed stacking ensemble model with the SMOTETomek outperforms the base learners and improves the accuracy of diagnosis and prediction of the MACE occurrences in patients with ACS at early stage.

INDEX TERMS Major adverse cardiovascular events (MACE), acute coronary syndrome (ACS), stacking ensemble classifier, machine learning, data sampling, imbalanced data.

I. INTRODUCTION

Cardiovascular disease (CVD) is the topmost cause of death worldwide. The World Health Organization (WHO) demonstrated that an estimated 17.9 million people died from CVDs in 2016, representing 31% of all global deaths [1]. Acute

coronary syndrome (ACS), a common and serious type of CVD, causes a lack of oxygen to the heart and can lead to unstable angina or myocardial infarction (MI) [2]. The prediction of major adverse cardiovascular events (MACE), as a critical tool to estimate the likelihood of the risk of ACS, has been widely used in the prevention and intervention of ACS at early stage [3], [4]. It includes several types of cardiac events, such as MI, coronary artery bypass grafting (CABG),

The associate editor coordinating the review of this manuscript and approving it for publication was Xiao-Sheng Si^{ID}.

re-percutaneous coronary intervention (re-PCI), cardiac death, and non-cardiac death, etc. So far, the History, Electrocardiogram (ECG), Age, Risk factors and Troponin (HEART), Thrombolysis in Myocardial Infarction (TIMI), Global Registry of Acute Coronary Events (GRACE) [5], Framingham risk scores [6], and other different kinds of risk scoring tools have been used to estimate the seriousness for ACS patients. Nevertheless, there are several limitations of those risk scores, which consider only very few patients' prognostic factors and have been generated long time ago. On the other hand, the previous traditional risk scoring tools cannot make a good prediction result for patients with ACS [5], [7]. Hence it is necessary to consider the new prognostic factors for prediction in patients with ACS over time.

For this reason, artificial intelligence (AI) techniques such as machine learning (ML) are recently focused and can not only enhance the performance while discovering multiple important prognostic factors, but also overcome some typical data problems such as missing values and outliers. Many different ML algorithms, such as logistic regression (LR), support vector machine (SVM), K-nearest neighbors (KNN), decision tree (DT), random forest (RF), extreme gradient boosting (XGBoost), and adaptive boosting (AdaBoost), have shown their powerful abilities in various domains [8]–[11]. Even though most of the individual ML-based prediction models of the MACE occurrences have outstanding performance, there are also several challenging problems as follows. First, no individual ML algorithm is always better than other one in the same domain. Second, the combination of different algorithms may get better performance than an individual algorithm [12]. Especially in medical domain where precise results are needed, the accurate result means the efficiency of diagnosis or the survival rate for patients. Third, there is an imbalanced problem among the classes of experimental dataset. So, the data sampling techniques need to be investigated to solve the bias issue of the class distribution [10], [13], [14].

Therefore, this paper proposes a stacking ensemble model for the MACE occurrences in patients with ACS using seven popular ML algorithms such as LR, SVM, KNN, DT, RF, XGBoost, and AdaBoost as base learners while dealing with the class imbalance problem through data sampling techniques. Our major contributions of this paper can be summarized as follows. First, the statistics-based multicollinearity analysis is used to verify the high correlation problem among all variables in experimental data. Second, we use three types of data sampling techniques, such as borderline synthetic minority oversampling technique (Borderline-SMOTE1), Cluster Centroids undersampling technique, and synthetic minority oversampling technique (SMOTE) plus Tomek Links (SMOTETomek) hybrid technology, to efficiently handle the class imbalance problem in experimental dataset and compare their performances on the balanced data with those on the original imbalanced data. Third, we discover the important prognostic factors

by applying the RF, XGBoost, and AdaBoost algorithms which can get higher performance than other base learners for the prediction of MACE occurrences in patients with ACS. Fourth, the KNN-based imputation approach, which has shown the efficiency to handle the missing values [15]–[17], is applied to solve the missing values when preprocessing the raw dataset. Finally, we propose a stacking ensemble prediction model using seven ML algorithms as base learners to overcome the limitations and improve the performance of the individual ML-based prediction models.

The structure of the paper proceeds as follows: Section II summarizes the related work about the prediction of MACE occurrences and imbalanced data solution in the medical domain. Section III describes the data and methods used in this paper. In Section IV, the experimental results are illustrated. And the discussion is summarized in Section V. Finally, the conclusion is given in Section VI.

II. RELATED WORK

A. THE PREDICTION OF MACE OCCURRENCES

In recent years, the death ratio of ACS is increasing very fast globally. Therefore, there has been an increasing interest in the prediction of MACE occurrences as it has a significant impact on medical decision-making for patient care and treatment.

In a study, a ML-based 1-year mortality prediction model after discharge in 8,227 patients with ACS were designed using gradient boosting machine, generalized linear model, RF, and Deep Neural Network [11]. In another study, a hybrid method with the ML-based models, i.e., SVM, LR, and classification and regression tree, was presented to generate the MACE prediction models in 2,930 patients with ACS [4]. In a similar research, a hybrid approach for MACE prediction in patients with ACS was developed by utilizing Chinese admission records including 2,930 ACS patient samples [18]. They applied four representative ML algorithms such as SVM, RF, naïve Bayes, and LR to construct MACE prediction models. In [5] the authors compared the performance of the GRACE, HEART, and TIMI scores in predicting the probability of MACE in 1,748 chest pain patients presenting at the emergency department. In [19] the authors constructed, validated, and measured the performance of MACE prediction within 1-year follow-up after percutaneous coronary intervention (PCI) based on three kinds of ML algorithms such as LR, artificial neural network, and DT.

Few ensemble learning methods were also used to predict the MACE occurrences. In a study, the authors developed a stacked ensemble ML prognostic model to predict the 60-day risk of MACE in adults with chest pain and compared the performance with different algorithms, such as lasso, splines, RF, XGBoost, Bayesian additive regression trees [20]. Another study used a boosted regularized multi-label learning approach to predict multi-type MACE after ACS [21].

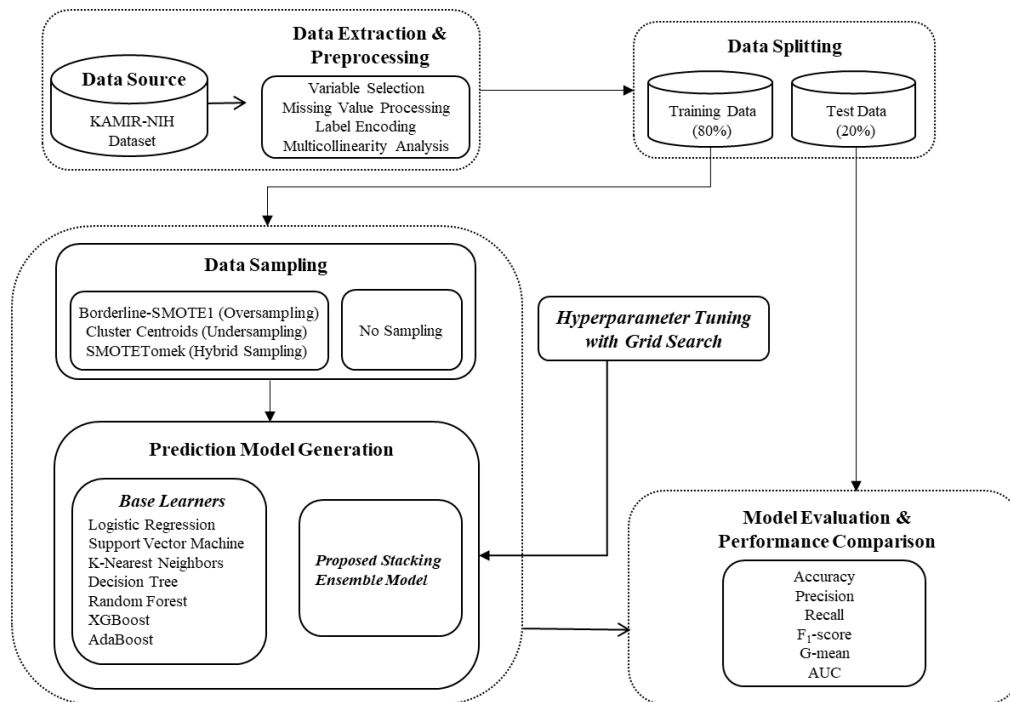


FIGURE 1. Overall experimental workflow for the prediction model of MACE occurrences. *Note: G-mean denotes geometric mean, AUC = area under the ROC curve.

B. IMBALANCED DATA SOLUTION IN MEDICAL DOMAIN

The class imbalance problem is very common in the medical domain and has received significant attention. The under-sampling and oversampling methods, such as random under-sampling, edited nearest neighbors, and SMOTE were used to handle the class imbalance on the osteoporosis dataset with several ML algorithms to generate the classification models [22]. Their outcomes showed that the highest efficiency was achieved while using the SMOTE and RF classifier. In addition, a study used the SMOTE with several ML algorithms to predict acute myocardial infarction <1 month and all-cause mortality <1 month for MACE in emergency department patients with chest pain [23]. Furthermore, in [24] the authors proposed a prediction algorithm for diabetes mellitus classification on imbalanced data using the adaptive synthetic (ADASYN) sampling technique to reduce the influence of class imbalance, then using a RF classifier to generate predictions models.

III. MATERIALS AND METHODS

A. OVERALL EXPERIMENTAL WORKFLOW

The overall experimental workflow for constructing the proposed stacking ensemble prediction model for the occurrences of MACE is shown in Fig. 1. First, our experimental data is extracted from the Korea Acute Myocardial Infarction Registry National Institutes of Health (KAMIR-NIH) dataset and preprocessed the extracted data, i.e., handling irrelevant variables and missing values, encoding some categorical variables, and we apply the multicollinearity analysis to solve

the high correlation problem of all variables [25]. Second, the experimental dataset is split into training (80%) and test data (20%) in a stratified fashion while considering the percentage of each class for further experiment. However, there is a class imbalance problem of experimental data after extraction. Third, we use three data sampling techniques, such as Borderline-SMOTE1 oversampling, Cluster Centroids undersampling, and SMOTETomek hybrid sampling approach, to deal with the class imbalance problem. Then, we develop a stacking ensemble prediction model for the MACE occurrences in patients with ACS using seven widely used ML algorithms as base learners and conduct the hyperparameter tuning for applied algorithms. Lastly, we evaluate the performance of the proposed stacking ensemble prediction model and compare with the seven base learners.

B. DATA SOURCE AND DATA EXTRACTION

For the experiment, we use the KAMIR-NIH dataset, which is the nationwide, multi-center registry of patients with cardiovascular disease in South Korea. The data was collected by the KAMIR from 52 different Korean community and university hospitals from November 2011 to December 2019 [26]. The raw dataset contains 13,104 patients' medical records with complete 2-year follow-ups and 550 variables of the patients with ACS. In this paper, the MACE is defined as the events such as MI, CABG, re-PCI, cardiac death, and non-cardiac death as the target variable.

The KAMIR-NIH dataset includes numerous irrelevant variables and plenty of missing values, so we extract and

preprocess the raw dataset before generating the prediction model. First, we eliminate the irrelevant variables which have no impact on the prediction of MACE occurrences [11], [27]. Next, the target variable is transformed into ‘No_MACE’ and ‘MACE’, where the MACE is defined as the prescribed five cardiac events. The values of all categorical variables are also converted into the integer values. For example, the value ‘Yes/No’ is converted into ‘1/0’. The applied variables extracted from the KAMIR-NIH dataset are shown in Table 1.

TABLE 1. The applied variables from the KAMIR-NIH dataset.

Type	Variables
Categorical (32)	Gender, Chest Pain, Dyspnea, Previous Chest Pain, Killip Class, Use of ECG at admission, ST-change on ECG, Hypertension, Diabetes Mellitus, Dyslipidemia, Previous MI, Previous Angina Pectoris, Previous Heart Failure, Previous Cerebrovascular Disease, History of Smoking, Family History of Heart Disease, Family History of Early Age Ischemic Heart Disease, Symptoms of MI, MI ECG Change, PCI, Pre-TIMI Flow of Target Vessel, Post-TIMI Flow of Target Vessel, Initial Diagnosis of STEMI & NSTEMI, Use of Thrombolysis, Outcome of thrombolysis, Use of CAG, Result of CAG, Use of ECG in hospital, CABG, Final Diagnosis of STEMI & NSTEMI, Discharge Type of Patient, MACE or No_MACE
Continuous (28)	Age, SBP, DBP, Heart Rate, Height, Weight, Abdominal circumference, WBC, WBC Neutrophil, WBC Lymphocyte, Hemoglobin, Platelet, Glucose, Creatinine, Maximum Creatine Kinase Peek, Maximum Creatine Kinase MB, Tnl, Total Cholesterol, Triglyceride, HDL, LDL, hsCRP, NTproBNP, BNP, HbA1c, Discharge SBP, Discharge DBP, Discharge Heart Rate

***Note:** ECG denotes Electrocardiogram, MI = Myocardial Infarction, PCI = Percutaneous Coronary Intervention, TIMI = Thrombolysis In Myocardial Infarction, STEMI = ST-segment elevation myocardial infarction, NSTEMI = Non-ST segment elevation myocardial infarction, CAG = Use of Coronary Angiogram, CABG = Coronary Artery Bypass Grafting, SBP = Systolic Blood Pressure, DBP = Diastolic Blood Pressure, WBC = White Blood Cells, Tnl = Maximum Troponin I, HDL = High-Density Lipoprotein cholesterol, LDL = Low-Density Lipoprotein cholesterol, hsCRP = C-reactive protein, NTproBNP = N-terminal Prohormone of Brain Natriuretic Peptide, BNP = B-type Natriuretic Peptide, HbA1c = Glycated Hemoglobin.

Then we excluded the patients’ records who died in hospital during admission (Excluded N = 504) and failed 2-year follow-up (Excluded N = 1,411) and extract the records related to the CVD alive patients with 2-year follow-up (N = 11,189) from the total records (N = 13,104). The data extraction process is illustrated in Fig. 2, where the final dataset includes the MACE records of 612 (5.47%) and No_MACE 10,577 (94.53%). However, there are still amounts of missing values in the final dataset. To deal with the missing values, we apply the KNN-based imputation approach, which computes the estimated value from the k closest samples among the dataset, and it is an efficient approach for missing value imputation [15]–[17], where the k value is initialized as 5 in this paper.

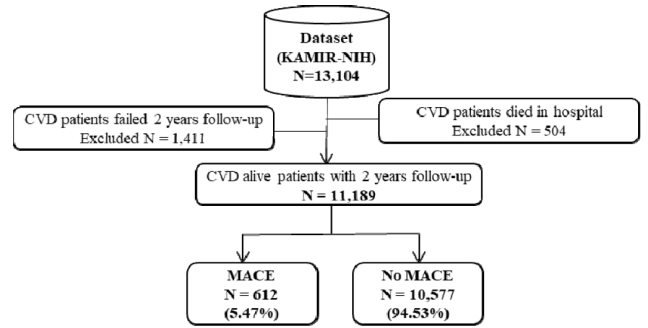


FIGURE 2. The data extraction process.

C. APPLIED DATA SAMPLING TECHNIQUES

Our final dataset shows a manifest class imbalance problem between MACE and No_MACE groups. Generally, the data sampling techniques can be classified into three categories: oversampling, undersampling and hybrid approaches [14], [28]. In this paper, three types of representative data sampling techniques, such as Borderline-SMOTE1 [29] oversampling, Cluster Centroids [30] undersampling, and SMOTETomek [31] hybrid technique, are applied to solve the class imbalance problem while training the prediction models.

1) BORDERLINE-SMOTE1 OVERSAMPLING

SMOTE, as the most common oversampling technique, has been proven highly effective in increasing the number of minority classes for getting balanced class distribution [32]. It adds new artificial synthetic minority class samples by extrapolating between pre-existing minority instances rather than simply random replacement of original samples, and it finds the k nearest neighbors of the minority class for each minority sample [33]. The k nearest neighbors from the KNN are randomly chosen depending on the level of oversampling required [32]. Among the SMOTE sampling techniques, this paper uses Borderline-SMOTE1 sampling and k-value as 5, which only oversamples the minority class and is very similar to SMOTE regular technique [29], [33].

2) CLUSTER CENTROIDS UNDERSAMPLING

Cluster Centroids is the representative undersampling technique to decrease the number of majority classes to achieve balanced class distribution. It performs undersampling by generating centroids based on clustering methods, which selects a representative subset from the majority class by replacing a cluster of majority samples by the cluster centroid of a K-means algorithm [30].

3) SMOTETOMEK HYBRID SAMPLING

SMOTETomek [31] is the typical hybrid sampling technique which is used to increase the number of minority class while decreasing the number of majority class to balance the class size. After using the SMOTE technique to add new artificial synthetic minority class samples, the Tomek Links [34]

technique is used to reduce the majority class data. Tomek Link represents the nearest pair of samples between different categories, that is, the two samples are nearest neighbors of each other and belong to different classes. Therefore, one from both samples is either noise or both samples are near to the boundary of the classes [35]. After that, cleaning up the overlap between classes by removing the Tomek Link can make all the nearest neighbor samples belong to the same class and be better classified.

D. APPLIED MACHINE LEARNING MODELS

In this paper, several widely used ML algorithms, such as LR [36], SVM [37], [38], KNN [39], DT [40], RF [41], XGBoost [42], and AdaBoost [43] are applied as base learners because of its extraordinary performance. It can also be used to predict the results from different aspects to increase the robustness of our proposed stacking ensemble model. A brief introduction of those ML algorithms is described as follows.

LR [36] is a commonly used linear model for classification rather than regression. It uses the sigmoid function to deal with the binary classification of variables by squashing the value within the range of [0,1].

SVM [37] is a powerful supervised learning algorithm for classification and regression analysis. The original idea of SVM is to construct a decision boundary named as hyperplane while mapping the data into a higher dimensional input space through either linear or non-linear kernel functions. Then, it uses the separating hyperplane to create a classifier by maximizing the distance between two classes. Consequently, this algorithm is also known as 'hyperplane classifier'. In this paper, we used the linear support vector classification (LinearSVC) to implement our experiment instead of the traditional SVM with the kernel because of its flexibility to large number of samples and is much faster than the SVM with kernel [38].

KNN [39] is a simple supervised learning approach and can be used to deal with both classification and regression problems. It is also a type of instance-based learning approach and works by the distance function. A sample is classified by the majority vote of its neighbors, and the sample is assigned to the most common type of its k nearest neighbors measured by the distance function, where the number of the k is an integer value specified by the user and the optimal choice of the value is highly data-dependent.

DT [40] is a non-parametric supervised learning approach used for classification and regression. The goal of the algorithm is to create a model for the target variable values predicted by learning simple decision rules inferred from several input variables. Amongst all ML algorithms, DT is very efficient and performs well in large datasets. It is considerably simple to understand and interpret.

RF [41] is very efficient in both classification and regression. It uses the DT algorithm as the base classifier with a bagging approach to generate multiple small decision trees by random sampling with replacement. Each tree in the RF

produces a class prediction model, and the class with the largest number of votes becomes the final prediction result of the model.

XGBoost [42] is an extension of Gradient Boosting Decision Tree designed to be highly efficient, flexible, and portable to solve both classification and regression problems. It is designed based on the boosting approach using the gradient descent algorithm to minimize the loss while adding the new models in the boosting technique.

AdaBoost [43] is useful in solving both classification and regression problems. It is called adaptive because it generates the strong learner while using multiple iterations and adds those weak learners iteratively to create the strong learner. In each round of training, a new weak learner is added to the ensemble process and the weight vector is adjusted to concentrate on the previous case of classification errors.

E. PROPOSED STACKING ENSEMBLE MODEL

The stacking technique is a meta-learning approach that builds its model using the predictions of multiple base learners as input to train a new meta learner, and then make the final prediction. It has been proven to be more effective than any individual algorithm in both classification and regression problems. In this paper, seven frequently used ML algorithms, such as LR, SVM, KNN, DT, RF, XGBoost, and AdaBoost are applied as base learners, then the RF algorithm is used as a meta learner to generate the proposed stacking ensemble prediction model for the prediction and diagnosis of MACE occurrences in patients with ACS at early stage since the RF can get a great result most of the time, which is shown in Fig. 3.

F. HYPERPARAMETER TUNING

To develop the accurate prediction model and minimize the classification error for the occurrences of MACE in patients with ACS, we optimize several important parameters of applied ML algorithms using the grid search approach. Table 2 shows the hyperparameter tuning of the applied ML algorithms. To reduce the calculation time for obtaining the optimal hyperparameters, the ranges of the hyperparameter configuration in each algorithm were tested with less parameters first and then set more parameters nearby the values obtained from the previous step to get the optimal ones. For example, the number of neighbors k in the KNN algorithm is initially searched from 1, 5, 9, 15, 19, because the k is recommended to take an odd k for binary problem [44], [45], if we get the optimal value as 15, then we set the values nearby this value as 13, 17 to search more accurate hyperparameters.

G. STATISTICAL ANALYSIS AND IMPLEMENTATION ENVIRONMENTS

For the statistical analysis of experimental data, we use the Chi-square test [46] and independent t-test [47] to analyze the categorical and continuous variables, respectively. The categorical variables are represented as frequency and its

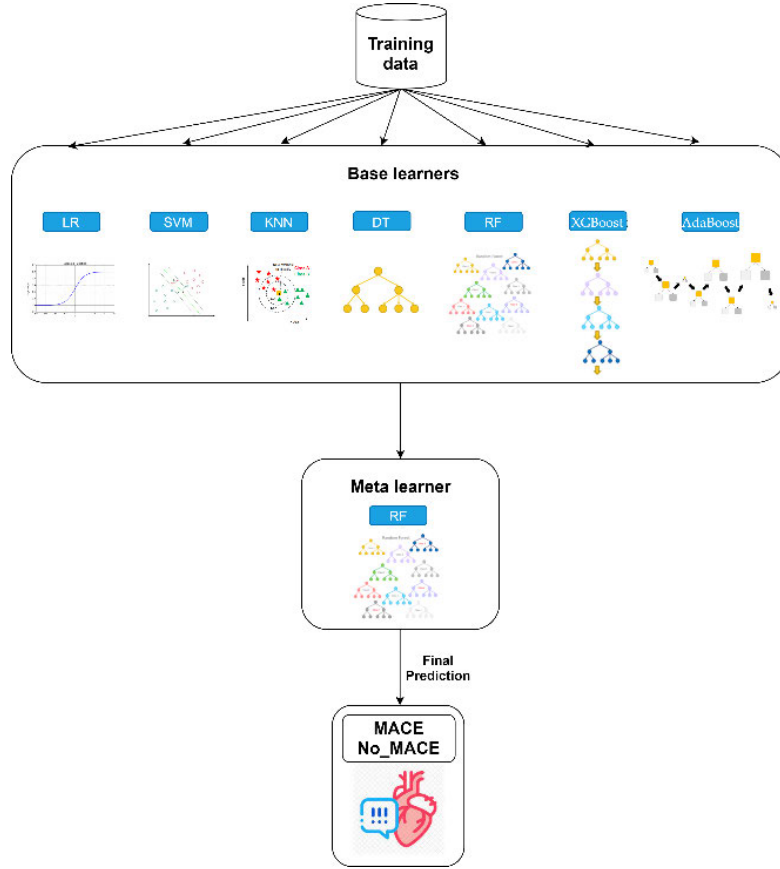


FIGURE 3. The proposed stacking ensemble prediction model for the occurrences of MACE.

TABLE 2. Hyperparameter tuning of machine learning algorithms with grid search approach.

Algorithm	Parameter Configuration	Range
LR	penalty solver	{‘l1’, ‘l2’, ‘elasticnet’, ‘none’} {‘newton-cg’, ‘lbfgs’, ‘liblinear’, ‘sag’, ‘saga’}
	c	{1e-5, 1e-1, 1, 100, 200, 300, 400, 500, 700, 900, 1000}
DT	criterion	{‘gini’ and ‘entropy’}
	number of estimators	{200, 500, 600, 700, 800, 900, 1000}
XGBoost	criterion	{‘gini’ and ‘entropy’}
	number of estimators	{100, 300, 500, 600, 700, 800, 900, 1000}
	eta (learning rate)	{0.001, 0.01, 0.1, 0.15, 0.2, 0.3}
	maximum depth	{3, 5, 7, 9}
KNN	gamma	{0, 0.1, 0.2, 0.3, 0.4, 0.5}
	number of neighbors	{1, 3, 5, 7, 9, 13, 15, 17, 19}
AdaBoost	number of estimators	{200, 400, 450, 600, 700, 800, 900, 1000}
	learning rate	{0.001, 0.01, 0.05, 0.1, 0.2, 0.3}

rate while continuous variables are represented as *mean* $\pm$ *standard deviation*. Furthermore, the multicollinearity analysis is applied to solve the correlation problem and test the multicollinearity of all independent variables. The variance inflation factor (VIF), which is an indicator of linear

dependence between variables, is used to indicate the multicollinearity in regression analysis [25]. The VIF can be calculated as

$$VIF = \frac{1}{1 - R^2} = \frac{1}{Tolerance} \quad (1)$$

where R^2 is the coefficient of determination and tolerance is the inverse of the VIF. More large VIF demonstrates the multicollinearity and practical experience indicates that if any of the VIFs exceeds 5 or 10, the associated regression coefficients are poorly estimated because of the multicollinearity [25]. The threshold of VIF is set to 5 in this paper.

All experiments were implemented on a PC with Intel Core i7 2.70 GHz, and 16GB random access memory (RAM) using the Microsoft Windows 10 operating system. To carry out the experiments, all statistical analysis were implemented using the IBM SPSS Statistics 23, the Python (Version 3.6) fancy-impute library to fill in the missing values, the imbalanced-learn library to solve the class imbalance problem [48], the scikit-learn package and XGBoost library to generate the ML-based prediction models.

H. PERFORMANCE EVALUATION

To evaluate the performance of ML-based prediction models for the MACE occurrences in patients with ACS,

the accuracy, precision, recall, F1-score, geometric mean (g-mean), and AUC are used to compare the performance of all applied prediction models. The equations of performance measures are shown in equations (2) to (6), where TP, TN, FP, and FN are the acronyms of true positive, true negative, false positive, and false negative in the confusion matrix, respectively.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-score} = \frac{(2 * \text{Recall} * \text{Precision})}{\text{Recall} + \text{Precision}} \quad (5)$$

$$\begin{aligned} \text{Geometric mean} &= \sqrt{\text{Recall} * \text{Specificity}} \\ &= \sqrt{\frac{TP}{TP + FN} * \frac{TN}{FP + TN}} \end{aligned} \quad (6)$$

IV. EXPERIMENTAL RESULTS

A. CLINICAL CHARACTERISTICS

In this paper, we extracted 11,189 ACS patients' records with 2-year follow-up as the experimental dataset which included the *MACE* records of 612 (5.47%) and *No_MACE* 10,577 (94.53%). The clinical characteristics between *MACE* and *No_MACE* groups in the experimental dataset were summarized in Table 3. The variables with significant differences between the two groups are shown as bold. Table 3 showed that among the *MACE* and *No_MACE* groups, the variables of *age*, *heart rate*, *height*, *weight*, *killip class*, *chest pain (typical)*, *dyspnea (yes)*, *ST-change on ECG (yes)*, *current smoker (yes)*, *PCI (yes)*, *history of hypertension (yes)*, *history of diabetes mellitus (yes)*, *history of dyslipidemia (yes)*, *previous myocardial infarction (yes)*, *previous angina pectoris (yes)*, *previous heart failure (yes)*, *previous cerebrovascular disease (yes)*, *pre-TIMI flow of target vessel*, *post-TIMI flow of target vessel*, *discharge type of patient (recovery)*, *neutrophil*, *lymphocyte*, *hemoglobin*, *platelet*, *glucose*, *creatinine*, *total cholesterol*, *LDL*, *hsCRP*, *NTproBNP*, *BNP*, and *HbA1c*, were statistically highly significant with $p\text{-value} \leq 0.001$ and *CABG (Yes)* 0.005, *triglyceride* 0.005, *discharge heart rate* 0.005, *gender* 0.01, *use of CAG (yes)* 0.05, *final diagnosis of STEMI & NSTEMI (STEMI)* 0.05, *discharge DBP* 0.05, *HDL* 0.05, respectively. In contrast, *abdominal circumference*, *SBP*, *DBP*, *previous chest pain (yes)*, *ECG (yes)*, *family history of heart disease (yes)*, *family history of early age ischemic heart disease (yes)*, *thrombolysis (yes)*, *outcome of thrombolysis (success)*, *result of CAG (success)*, *use of ECG in hospital (yes)*, *discharge SBP*, *WBC*, *platelet*, *maximum Creatine Kinase Peek*, *maximum Creatine Kinase MB*, and *TnI*, are statistically less significant.

B. RESULTS OF MULTICOLLINEARITY

Table 4 showed the VIF values of all variables in the experimental dataset. In Table 4, 11 variables showed higher VIF

values than the given threshold with multicollinearity problem (e.g. *previous cerebrovascular disease*, *family history of heart disease*, etc.). Therefore, those variables were excluded for our experiment.

C. RESULTS OF PREDICTION MODELS

In this section, the results of various ML-based prediction models were shown for the prediction of MACE occurrences in patients with ACS on imbalanced and balanced datasets. To generate the balanced data, we used three types of data sampling techniques, such as Borderline-SMOTE1 oversampling, Cluster Centroids undersampling, and SMOTETomek hybrid technique. The performance of the proposed stacking ensemble prediction model was evaluated and compared with base learners with respect to accuracy, precision, recall, F1-score, g-mean, and AUC on imbalanced and balanced datasets, as shown in Tables 5-8. The best results for each performance measure are shown in bold in each table.

Table 5 showed the performance evaluation of the proposed stacking ensemble and other seven base learners LR, SVM, KNN, DT, RF, XGBoost, and AdaBoost-based prediction models for the MACE occurrences on imbalanced data. The result showed that those models can get more than accuracy of 94% in all prediction models except the DT, but the precision, recall, F1-score, g-mean, and AUC on imbalanced data were very low in all prediction models. It means that those models get the biased result toward the majority class and cannot predict the minority class accurately.

Tables 6-8 compared the performance of the proposed stacking ensemble and other seven prediction models on balanced data with the oversampling, undersampling and hybrid sampling techniques, respectively. The proposed stacking ensemble prediction model showed the best performance with accuracy of 0.9839, precision 0.9961, F1-score 0.9837, g-mean 0.9838, and AUC 0.9839 on balanced data with oversampling technique (Table 6). It also showed the best performance with accuracy of 0.9862, F1-score 0.9862, g-mean 0.9863, and AUC 0.9863 on balanced data with hybrid sampling technique (Table 8). On the other hand, the RF showed the best performance on balanced data with undersampling technique with accuracy of 0.9796, recall 1, F1-score 0.9789, g-mean 0.9804, and AUC 0.9806 (Table 7).

Additionally, Fig. 4 showed the top 5 most important prognostic factors of base learners (e.g., RF, XGBoost, and AdaBoost) for the prediction of MACE occurrences with respect to the oversampling and hybrid techniques, because they got more accurate results than other base learners. In details, Fig. 4(a)-(c) showed the feature importance of RF, XGBoost, and AdaBoost using the oversampling technique, where variables *DBP*, *discharge heart rate*, *total cholesterol*, *abdominal circumference*, etc. showed their importance in those models. Fig. 4(d)-(f) showed the feature importance of RF, XGBoost, and AdaBoost using the hybrid sampling technique, where variables *platelet*, *TnI*, *triglyceride* etc. showed the importance in those models. Note that the stacking ensemble classifier has no feature importance attribute.

TABLE 3. The clinical characteristics of experimental dataset.

Variable	All (N=11189)	No_MACE (N=10577)	MACE (N=612)	p-value
Baseline clinical characteristics				
Gender				$\leq 0.01^{**}$
Male	8406 (75.1)	7977 (75.4)	429 (70.1)	
Female	2783 (24.9)	2600 (24.6)	183 (29.9)	
Age (years)	62.92 $\pm$ 12.29	62.64 $\pm$ 12.24	67.83 $\pm$ 12.04	$\leq 0.001^{***}$
Heart Rate (bpm)	77.83 $\pm$ 18.50	77.57 $\pm$ 18.32	82.42 $\pm$ 20.89	$\leq 0.001^{***}$
Height (cm)	164.55 $\pm$ 10.64	164.63 $\pm$ 10.74	163.16 $\pm$ 8.63	$\leq 0.001^{***}$
Weight (kg)	65.89 $\pm$ 11.92	66.06 $\pm$ 11.92	62.96 $\pm$ 11.61	$\leq 0.001^{***}$
Abdominal Circumference (cm)	87.10 $\pm$ 6.17	87.1 $\pm$ 6.14	86.95 $\pm$ 6.70	0.562
Killip Class				$\leq 0.001^{***}$
I	9124 (81.5)	8702 (82.3)	422 (69.0)	
II	923 (8.2)	855 (8.1)	68 (11.1)	
III	684 (6.1)	597 (5.6)	87 (14.2)	
IV	458 (4.1)	423 (4.0)	35 (5.7)	
SBP (mmHg)	131.71 $\pm$ 28.73	131.78 $\pm$ 28.65	130.48 $\pm$ 29.99	0.278
DBP (mmHg)	79.64 $\pm$ 17.45	79.71 $\pm$ 17.44	78.52 $\pm$ 17.67	0.101
Chest Pain (Typical)	9873 (88.2)	9379 (88.7)	494 (80.7)	$\leq 0.001^{***}$
Dyspnea (Yes)	2438 (21.8)	2265 (21.4)	173 (28.3)	$\leq 0.001^{***}$
Previous Chest Pain (Yes)	2903 (25.9)	2758 (26.1)	145 (23.7)	0.191
ECG (Yes)	11167 (99.8)	10556 (99.8)	611 (99.8)	0.849
ST-change on ECG (Yes)	8690 (77.7)	8205 (77.6)	485 (79.2)	$\leq 0.001^{***}$
Symptoms of MI (Yes)	11007 (98.4)	10408 (98.4)	599 (97.9)	0.317
MI ECG Change (Yes)	7561 (67.6)	7135 (67.5)	426 (69.6)	0.269
Current Smoker (Yes)	269 (2.4)	255 (2.4)	14 (2.3)	$\leq 0.001^{***}$
PCI (Yes)	10187 (91.0)	9652 (91.3)	535 (87.4)	$\leq 0.001^{***}$
Risk Factors				
History of Hypertension (Yes)	5603 (50.1)	5235 (49.5)	368 (60.1)	$\leq 0.001^{***}$
History of Diabetes Mellitus (Yes)	3042 (27.2)	2797 (26.4)	245 (40.0)	$\leq 0.001^{***}$
History of Dyslipidemia (Yes)	1325 (11.8)	1258 (11.9)	67 (10.9)	$\leq 0.001^{***}$
Previous Myocardial Infarction (Yes)	815 (7.3)	736 (7.0)	79 (12.9)	$\leq 0.001^{***}$
Previous Angina Pectoris (Yes)	1050 (9.4)	956 (9.0)	94 (15.4)	$\leq 0.001^{***}$
Previous Heart Failure (Yes)	141 (1.3)	114 (1.1)	27 (4.4)	$\leq 0.001^{***}$
Previous Cerebrovascular Disease (Yes)	690 (6.2)	628 (5.9)	62 (10.1)	$\leq 0.001^{***}$
Family History of Heart disease (Yes)	739 (6.6)	706 (6.7)	33 (5.4)	0.075
Family History of Early Age Ischemic Heart Disease (Yes)	83 (0.7)	80 (0.8)	3 (0.5)	0.433
Pre-TIMI Flow of Target Vessel	10201 (91.2)	9666 (91.4)	535 (87.4)	$\leq 0.001^{***}$
Post-TIMI Flow of Target Vessel	10201 (91.2)	9666 (91.4)	535 (87.4)	$\leq 0.001^{***}$
Initial Diagnosis of STEMI & NSTEMI (STEMI)	5819 (52.0)	5475 (51.8)	344 (56.2)	$\leq 0.05^*$
Thrombolysis (Yes)	117 (1.0)	110 (1.0)	7 (1.1)	0.806
Outcome of Thrombolysis (Success)	93 (0.8)	87 (0.8)	6 (1.0)	0.173
Use of CAG (Yes)	11092 (99.1)	10490 (99.2)	602 (98.4)	$\leq 0.05^*$
Result of CAG (Success)	11034 (98.6)	10433 (98.6)	601 (98.2)	0.088
Use of ECG in hospital (Yes)	10884 (97.3)	10294 (97.3)	590 (96.4)	0.175
CABG (Yes)	143 (1.3)	132 (1.2)	11 (1.8)	$\leq 0.005^{**}$
Final Diagnosis of STEMI & NSTEMI (STEMI)	5389 (48.2)	5122 (48.4)	267 (43.6)	$\leq 0.05^*$
Discharge Type of Patient (Recovery)	11020 (98.5)	10437 (98.7)	583 (95.3)	$\leq 0.001^{***}$
Discharge Heart Rate	70.65 $\pm$ 10.15	70.58 $\pm$ 10.10	71.86 $\pm$ 10.96	$\leq 0.005^{**}$
Discharge SBP (mmHg)	112.94 $\pm$ 14.62	112.88 $\pm$ 14.53	113.94 $\pm$ 16.11	0.081
Discharge DBP (mmHg)	68.10 $\pm$ 9.70	68.15 $\pm$ 9.68	67.30 $\pm$ 10.02	$\leq 0.05^*$
Laboratory findings				
WBC (103/mL)	10.31 $\pm$ 3.82	10.30 $\pm$ 3.80	10.41 $\pm$ 4.17	0.487
Neutrophil (%)	65.87 $\pm$ 14.96	65.72 $\pm$ 14.89	68.57 $\pm$ 15.86	$\leq 0.001^{***}$
Lymphocyte (%)	25.23 $\pm$ 12.89	25.38 $\pm$ 12.84	22.71 $\pm$ 13.45	$\leq 0.001^{***}$
Hemoglobin (g/dL)	13.95 $\pm$ 2.03	14.01 $\pm$ 1.99	12.99 $\pm$ 2.42	$\leq 0.001^{***}$
Platelet (mm)	232.62 $\pm$ 64.96	232.63 $\pm$ 64.15	232.52 $\pm$ 77.70	0.968
Glucose (mg/dL)	164.22 $\pm$ 73.71	163.23 $\pm$ 72.48	181.32 $\pm$ 90.77	$\leq 0.001^{***}$
Creatinine (mg/dL)	1.07 $\pm$ 1.12	1.04 $\pm$ 1.06	1.54 $\pm$ 1.80	$\leq 0.001^{***}$
Maximum Creatine Kinase Peek (mg/dL)	908.96 $\pm$ 1517.04	906.77 $\pm$ 1485.58	946.93 $\pm$ 1984.64	0.524
Maximum Creatine Kinase MB (mg/dL)	108.31 $\pm$ 157.41	109.00 $\pm$ 159.12	96.52 $\pm$ 123.79	0.057
TnI (mg/dL)	46.38 $\pm$ 96.94	46.38 $\pm$ 97.70	40.45 $\pm$ 82.90	0.985
Total Cholesterol (mg/dL)	179.56 $\pm$ 44.66	180.25 $\pm$ 44.59	167.67 $\pm$ 44.20	$\leq 0.001^{***}$
Triglyceride (mg/dL)	135.42 $\pm$ 118.50	136.18 $\pm$ 119.73	122.28 $\pm$ 93.74	$\leq 0.005^{**}$
HDL (mg/dL)	43.18 $\pm$ 12.06	43.24 $\pm$ 11.85	42.15 $\pm$ 15.12	$\leq 0.05^*$
LDL (mg/dL)	113.44 $\pm$ 37.74	114.04 $\pm$ 37.71	103.11 $\pm$ 36.72	$\leq 0.001^{***}$
hsCRP (mg/dL)	1.63 $\pm$ 4.95	1.54 $\pm$ 3.92	3.09 $\pm$ 13.43	$\leq 0.001^{***}$
NTproBNP (mg/dL)	1671.56 $\pm$ 4504.24	1515.32 $\pm$ 4065.63	4371.81 $\pm$ 8812.43	$\leq 0.001^{***}$
BNP (mg/dL)	306.80 $\pm$ 433.17	300.18 $\pm$ 424.60	421.31 $\pm$ 548.90	$\leq 0.001^{***}$
HbA1c (%)	6.34 $\pm$ 1.24	6.33 $\pm$ 1.23	6.55 $\pm$ 1.35	$\leq 0.001^{***}$

Note: The single asterisk (), the double asterisk (**), and triple asterisk (***) with p-value denote that those variables are statistically significant as p-value ≤ 0.05 , < 0.01 , and < 0.001 in *No_MACE* and *MACE* groups, respectively.

TABLE 4. Summary of regression coefficients.

Variable	Unstandardized Coefficients B	Standardized Coefficients Beta	t-value	p-value	Tolerance	VIF
Gender	0.017	.032	2.390	.017	.471	2.122
Age	0.001	.052	4.032	.000	.521	1.921
Chest Pain	-0.006	-.009	-.866	.386	.773	1.293
Dyspnea	-.005	-.009	-.906	.365	.882	1.134
Previous Chest Pain	-0.010	-.018	-1.954	.051	.965	1.037
SBP	-0.0000562	-.007	-.408	.683	.283	3.528
DBP	0.000	.020	1.179	.238	.286	3.497
Heart Rate	.000	.030	2.877	.004	.806	1.241
Killip Class	.009	.030	2.822	.005	.755	1.324
Height	.000	.009	.746	.456	.545	1.834
Weight	.000	-.019	-1.308	.191	.419	2.387
Abdominal Circumference	.000	.013	1.222	.222	.746	1.341
ECG	.009	.002	.187	.852	.988	1.012
ST-change on ECG	.000	-.013	-1.168	.243	.663	1.509
Hypertension	-.003	-.012	-.584	.559	.210	4.768
Diabetes Mellitus	.001	.004	.178	.859	.220	4.537
Dyslipidemia	-.009	-.025	-1.404	.160	.270	3.698
Previous MI	.024	.059	2.938	.003	.213	4.685
Previous Angina Pectoris	.020	.051	2.686	.007	.241	4.146
Previous Heart Failure	.043	.099	4.812	.000	.202	4.961
Previous Cerebrovascular Disease	-.017	-.139	-4.184	.000	.078	12.821
History of Smoking	.001	.003	.280	.780	.661	1.513
Family History of Heart Disease	-.013	-.029	-1.285	.199	.167	5.998
Family History of Early Age Ischemic Heart Disease	.026	.033	1.449	.147	.167	5.993
Symptoms of MI	.028	.015	1.548	.122	.863	1.159
MI ECG Change	.009	.019	1.439	.150	.510	1.962
WBC	.000	-.003	-.272	.785	.611	1.637
Neutrophil	.000	.019	.783	.433	.143	7.009
Lymphocyte	.000	.020	.801	.423	.142	7.027
Hemoglobin	-.005	-.043	-3.287	.001	.497	2.011
Platelet	0.00005308	.015	1.481	.139	.818	1.222
Glucose	0.00001293	.004	.330	.741	.533	1.876
Creatinine	.009	.046	4.500	.000	.804	1.243
Maximum Creatine Kinase Peek	0.00000351	.023	2.145	.032	.719	1.391
Maximum Creatine Kinase MB	-0.0000003101	.000	-.018	.985	.630	1.588
TnI	0.00002331	.010	.966	.334	.810	1.234
Total Cholesterol	.000	.025	1.206	.228	.203	4.931
Triglyceride	-0.00001352	-.007	-.606	.544	.635	1.575
HDL	.000	-.014	-1.318	.187	.781	1.280
LDL	.000	-.033	-1.772	.077	.248	4.026
hsCRP	.002	.036	3.661	.000	.897	1.115
NTproBNP	0.000004041	.080	7.441	.000	.741	1.350
BNP	-0.0000007233	-.001	-.132	.895	.783	1.278
HbA1c	.004	.021	1.618	.106	.509	1.963
PCI	-.035	-.043	-1.054	.292	.050	19.837
Pre-TIMI Flow of Target Vessel	.005	.028	2.442	.015	.632	1.582
Post-TIMI Flow of Target Vessel	-.013	-.067	-1.630	.103	.050	19.872
Initial Diagnosis of STEMI & NSTEMI	.022	.049	.976	.329	.035	28.887
Use of Thrombolysis	.084	.037	1.674	.094	.171	5.843
Outcome of thrombolysis	-.039	-.033	-1.488	.137	.172	5.806
Use of CAG	.017	.007	.619	.536	.662	1.509
Result of CAG	.003	.009	.776	.438	.599	1.670
Use of ECG in hospital	.016	.012	1.246	.213	.989	1.011
CABG	0.000004006	.000	.001	1.000	.842	1.188
Final Diagnosis of STEMI & NSTEMI	-.020	-.043	-.869	.385	.035	28.836
Discharge Type of Patient	.058	.032	3.383	.001	.962	1.039
Discharge SBP	.000	.011	.811	.417	.468	2.136
Discharge DBP	-.001	-.022	-1.733	.083	.511	1.958
Discharge Heart Rate	-0.00001862	-.001	-.085	.932	.907	1.102

Fig. 5 showed the normalized confusion matrices of the proposed stacking ensemble prediction models with (a) over-sampling, (b) undersampling, and (c) hybrid sampling techniques, where the predicted label on x-axis and true label

on y-axis means the predicted class labels and the actual class labels, as well as 0 and 1 indicates the *No_MACE* and *MACE* occurrences, respectively. The higher value of the coordinate (0, 0) and (1, 1) in the confusion matrix represents

TABLE 5. Performance evaluation of prediction models for MACE occurrences on imbalanced data.

Model	Accuracy	Precision	Recall	F1-score	G-mean	AUC
LR	0.9441	0.4	0.0161	0.031	0.1269	0.5074
SVM	0.9441	0	0	0	0	0.4998
KNN	0.9415	0	0	0	0	0.4983
DT	0.8937	0.087	0.0968	0.0916	0.3017	0.5186
RF	0.9441	0	0	0	0	0.4998
XGBoost	0.941	0	0	0	0	0.4981
AdaBoost	0.9433	0.2857	0.0161	0.0305	0.1268	0.5069
Stacking	0.9441	0.3333	0.0081	0.0157	0.0898	0.5036

TABLE 6. Performance evaluation of prediction models for MACE occurrence on balanced data with oversampling technique.

Model	Accuracy	Precision	Recall	F1-score	G-mean	AUC
LR	0.7258	0.7409	0.6924	0.7158	0.725	0.7257
SVM	0.7192	0.7347	0.6839	0.7084	0.7183	0.7191
KNN	0.8873	0.8257	0.981	0.8967	0.8826	0.8875
DT	0.9447	0.9396	0.9502	0.9449	0.9447	0.9447
RF	0.9742	0.996	0.9521	0.9736	0.9739	0.9742
XGBoost	0.9738	0.9955	0.9517	0.9731	0.9735	0.9737
AdaBoost	0.9636	0.9915	0.9351	0.9624	0.9631	0.9635
Stacking	0.9839	0.9961	0.9716	0.9837	0.9838	0.9839

TABLE 7. Performance evaluation of prediction models for MACE occurrences on balanced data with undersampling technique.

Model	Accuracy	Precision	Recall	F1-score	G-mean	AUC
LR	0.5755	0.5484	0.5862	0.5667	0.576	0.576
SVM	0.5918	0.5635	0.6121	0.5868	0.5925	0.5929
KNN	0.5633	0.529	0.7069	0.6052	0.554	0.5705
DT	0.9224	0.9292	0.9052	0.917	0.9214	0.9216
RF	0.9796	0.9587	1	0.9789	0.9804	0.9806
XGBoost	0.9796	0.9744	0.9828	0.9785	0.9797	0.9798
AdaBoost	0.9633	0.9652	0.9569	0.961	0.9629	0.9629
Stacking	0.9673	0.9737	0.9569	0.9652	0.9668	0.9668

TABLE 8. Performance evaluation of prediction models for MACE occurrences on balanced data with hybrid sampling technique.

Model	Accuracy	Precision	Recall	F1-score	G-mean	AUC
LR	0.6644	0.685	0.6156	0.6485	0.6628	0.6647
SVM	0.6582	0.678	0.61	0.6422	0.6567	0.6585
KNN	0.8254	0.7454	0.9915	0.851	0.8074	0.8245
DT	0.9368	0.9336	0.9411	0.9374	0.9367	0.9367
RF	0.9694	1	0.9392	0.9687	0.9691	0.9696
XGBoost	0.9694	0.9985	0.9407	0.9687	0.9692	0.9696
AdaBoost	0.9567	0.9964	0.9171	0.9551	0.956	0.9569
Stacking	0.9862	0.9976	0.975	0.9862	0.9863	0.9863

the higher accuracy of the prediction model for the *No_MACE* and *MACE* occurrences. The results showed that the proposed stacking ensemble prediction models got higher accuracy for both *No_MACE* and *MACE* occurrences with the data sampling techniques.

V. DISCUSSION

The prediction of MACE occurrences has a significant impact on medical decision making for the patient care and treatment. Achieving the accurate results for the prediction and diagnosis of MACE occurrences at early stage has been an important task in modern health care system since the development of new technology [4], [5], [11], [18], [19]. In this paper, we developed the prediction and diagnosis model for MACE occurrences in ACS patients based on the stacking ensemble approach. To generate the prediction model of MACE occurrences accurately, the statistics-based

multicollinearity analysis is used to test the correlation problem of all variables. Table 4 showed that 11 variables had the multicollinearity problems with higher VIF values. So, those variables were excluded from our experimental data.

The experiment was conducted by using the KAMIR-NIH data which was real-world ACS dataset with a 2-year follow-up. To overcome the significant class imbalance problem in our dataset, three types of principal Borderline-SMOTE1 oversampling, Cluster Centroids undersampling, and SMOTETomek hybrid data sampling techniques were used. Table 5 showed the results of prediction models for MACE occurrences in patients with ACS on original imbalanced dataset. Then, we found that all applied prediction models got biased results and low performance on the imbalanced data due to the lack of samples in the minority class. In contrast, the performance of all ML-based prediction

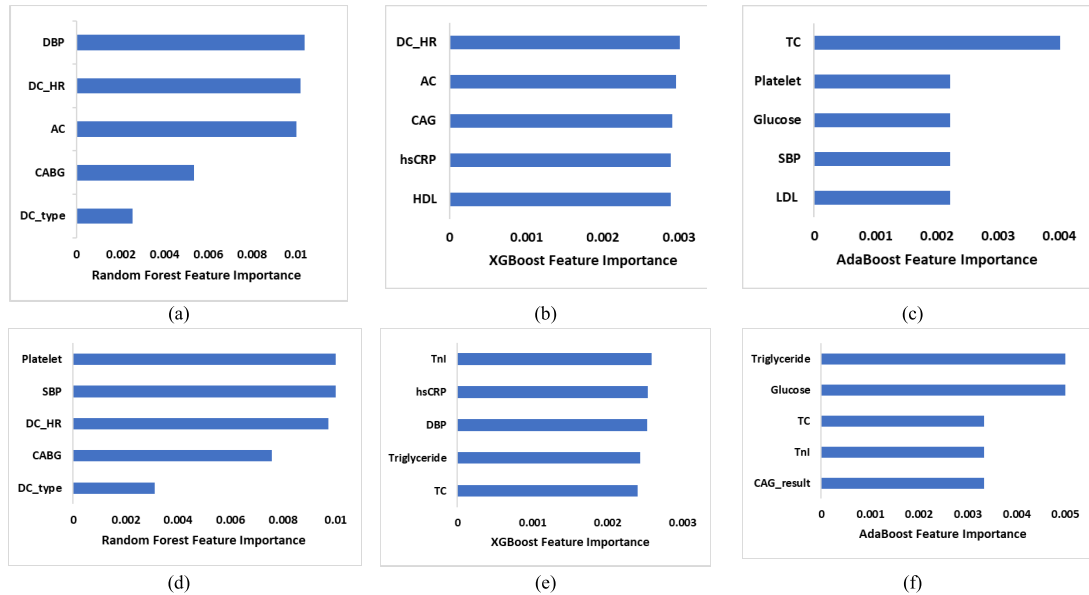


FIGURE 4. Top 5 primary prognostic factors of (a) Random Forest Prediction Model, (b) XGBoost, and (c) AdaBoost using oversampling; (d) Random Forest Prediction Model, (e) XGBoost, and (f) AdaBoost using hybrid sampling techniques.

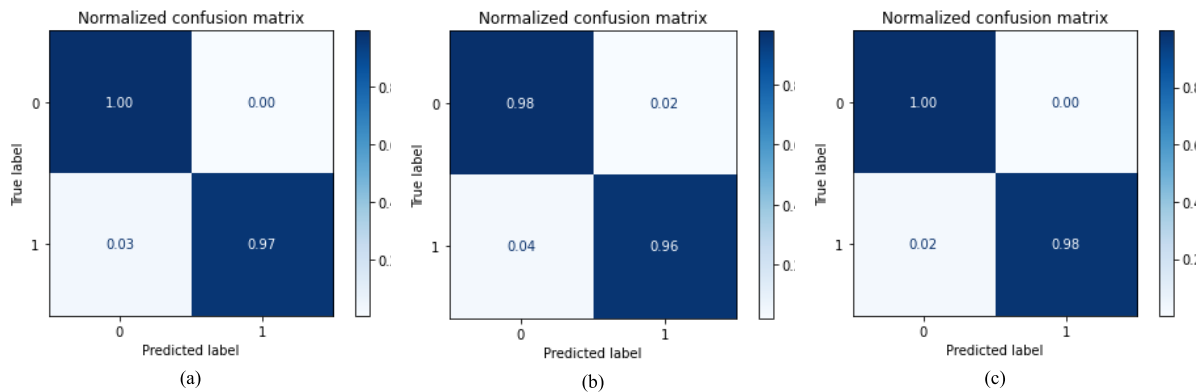


FIGURE 5. The normalized confusion matrices for the proposed stacking ensemble prediction models with the (a) oversampling; (b) undersampling; (c) hybrid sampling techniques.

models for the occurrences of MACE were significantly improved on the balanced data using data sampling techniques, as shown in Tables 6-8. Compared with the performances on original imbalanced data in Table 5, the results in Tables 6-8 showed the high improvement of precision, recall, F1-score, g-mean, and AUC in each ML-based prediction model on all balanced data with data sampling techniques.

In addition, the proposed stacking ensemble model was compared with other seven widely used ML-based prediction models such as LR, SVM, KNN, DT, RF, XGBoost, and AdaBoost. As the results shown in Tables 6-8, we found that our proposed stacking ensemble prediction model using the oversampling and hybrid sampling techniques improved the performance of applied base learners. In particular, the proposed stacking ensemble prediction model with the SMOTETomek hybrid sampling technique got the best performance measures with the accuracy of 0.9862, precision 0.9976, recall 0.975, F1-score 0.9862, g-mean 0.9863, and

AUC 0.9863 for the prediction and diagnosis of the MACE occurrences in patients with ACS, as shown in Table 8. There was one obvious weak point of the undersampling technique. That is, compared with the oversampling and hybrid sampling techniques, the undersampling technique could lose lots of useful information from the raw dataset. Furthermore, the experimental results demonstrated that the ensemble learning-based classifiers, such as RF, XGBoost, and AdaBoost got better performance than other base learners like LR, SVM, KNN, and DT. On the other hand, the RF also showed the highest performance with the oversampling, undersampling, and hybrid sampling techniques than other six ML-based prediction models. The normalized confusion matrices shown in Fig. 5 also indicated that the proposed stacking ensemble model provided higher performance for the prediction of MACE occurrences and gave a better solution on balanced data by the data sampling techniques.

Lastly, the ML-based prediction models could extract the hidden prognostic factors from the data automatically. Therefore, the top 5 primary prognostic factors of the RF, XGBoost, and AdaBoost-based prediction models were provided since they got the more accurate results than other base learners. The result from the Fig. 4 (a)-(f) showed that the prognostic factors of ML-based prediction models were different from those of the conventional risk scoring tools, such as HEART, TIMI, or GRACE [5].

However, this paper has several limitations. First, the paper was limited to the Korean population because we concentrated on the Korean patients' dataset with ACS for our experiments. So, it is difficult to extend the final result to other populations in the world. Second, our experimental dataset was only with 2-year follow-up, hence the prediction and diagnosis model for the MACE occurrences in patients with ACS may not be appropriate for in-hospital patients or patients with short-term follow-up. Third, the time and space complexities for ensemble learning model are higher than the individual classification models; moreover, it is difficult to interpret the ensemble learning model.

VI. CONCLUSION

This paper presented a stacking ensemble model for the prediction and diagnosis of the MACE occurrences at early stage in patients with ACS during 2-year follow-up using seven widely used ML-based classifiers. The prediction model also used oversampling, undersampling, and hybrid sampling techniques to deal with the class imbalance problem. Consequently, our findings were that the proposed stacking ensemble prediction model with the SMOTETomek hybrid sampling technique outperformed other ML-based models and the important prognostic factors for applied RF, XGBoost, and AdaBoost were different from the traditional statistical methods. Nowadays, the deep learning approach is popular, so we will focus on the development of deep learning-based prediction model for the prediction of MACE occurrences in the future.

ACKNOWLEDGMENT

The authors would like to thank the editor and the reviewers for their suggestions to improve the manuscript. They also like to thank the Korea Acute Myocardial Infarction Registry (KAMIR), a nationwide, multicenter data collection registry, to provide the data for the experiments.

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